

Quench Temperature During Reflood

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Abstract

There exist correlations to predict the quenching temperature (hereafter $T_{\min}$); however, when implemented in TRACE they led to incorrect predictions of the quench-front behaviour during a reflood experiment performed in the RBHT facility. This discrepancy suggests that the built-in $T_{\min}$ models are not representative of the conditions encountered in RBHT reflood tests. To address this issue, alternative correlations were reviewed from the literature and assessed for suitability. In addition, new $T_{\min}$ correlations were derived directly from experimental measurements. The development was based on data from three dedicated reflood experiments, used to identify the dominant trends and calibrate model parameters. The resulting correlations better reproduce the observed quench-front progression under the tested conditions. Overall, this work shows that the $T_{\min}$ correlation required for the ongoing experiment can differ substantially from those currently implemented in TRACE, and that facility- and condition-specific tuning may be necessary for reliable reflood predictions.

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1 Introduction

1.1 Context and motivation

The experiment consists of reflooding a bundle of heated rods and measuring the quench-front height as a function of time. The investigated experiments are three:

	Pressure [MPa]	Mass flux [$\text{kg s}^{-1} \text{m}^{-2}$]	T_{wall} [K]	T_{inlet} [K]
Case 9005	0.2648	52.1918	999.7	394.1
Case 9015	0.2730	161.0000	1144.1	391.9
Case 9021	0.2654	25.9885	999.7	393.6

Table 1: Operating conditions for the three cases.

The available correlations do not reproduce the experimental trends accurately when key parameters are varied, leading to significant discrepancies in the predicted quench-front progression.

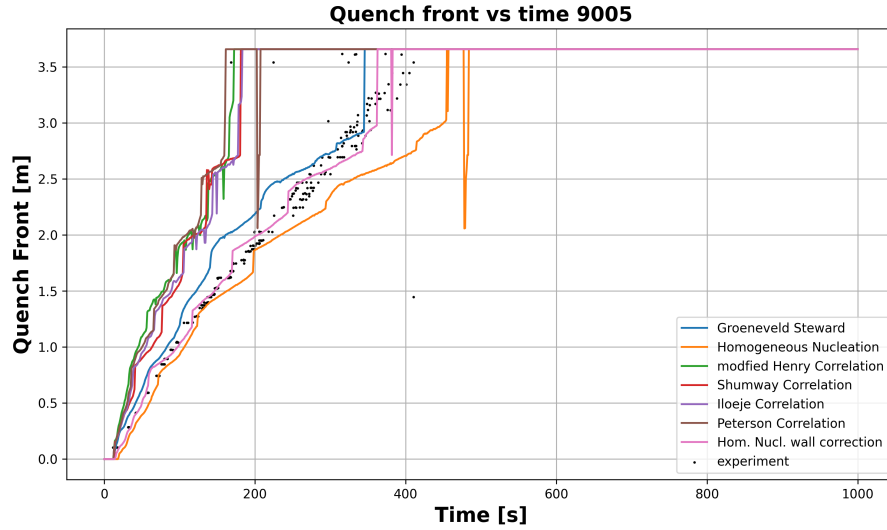


Figure 1: Quench front height vs time (case 9005).

	Pressure [MPa]	Mass flux [$\text{kg s}^{-1} \text{m}^{-2}$]	T_{wall} [K]	T_{inlet} [K]
Case 9005	0.2648	52.1918	999.7	394.1

Table 2: Operating conditions for case 9005.

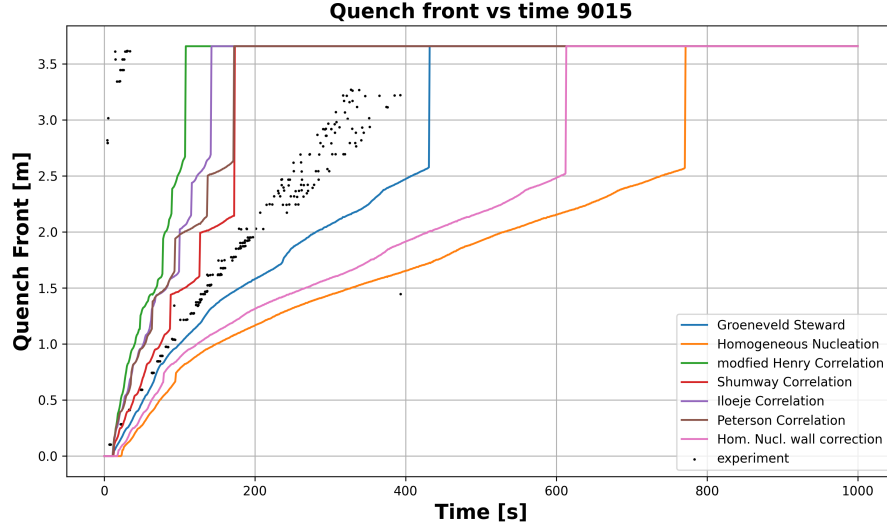


Figure 2: Quench front height vs time (case 9015).

	Pressure [MPa]	Mass flux [$\text{kg s}^{-1} \text{m}^{-2}$]	T_{wall} [K]	T_{inlet} [K]
Case 9015	0.2730	161.0000	1144.1	391.9

Table 3: Operating conditions for case 9015.

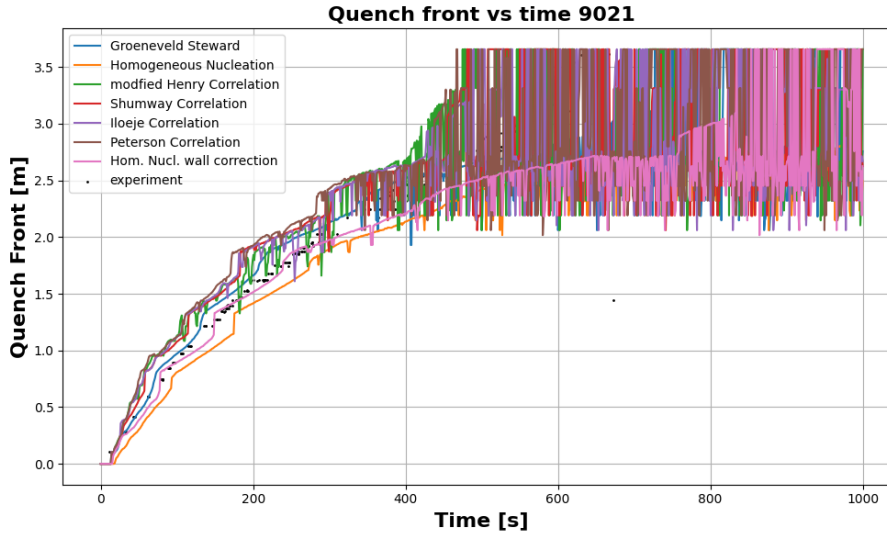


Figure 3: Quench front height vs time (case 9021).

	Pressure [MPa]	Mass flux [$\text{kg s}^{-1} \text{m}^{-2}$]	T_{wall} [K]	T_{inlet} [K]
Case 9021	0.2654	25.9885	999.7	393.6

Table 4: Operating conditions for case 9021.

It can be noticed that in the 9005 case two correlations: "Groeneveld Steward" and "Homogeneous Nucleation with Wall Correction" well predict the experimental data but as we change the

mass flux, increasing it, as in 9015 case, there is no correlation that well predict the experimental data.

2 Background and Related Work

2.1 Quenching and definition of $T_{\min}$

During reflood, the liquid level is typically higher than the quench front; in fact, a stable vapour film insulate the rod surface and prevent direct contact between the bulk water and the cladding. The quench front marks the axial location where this insulating film collapses and efficient cooling by liquid rewetting is re-established.

To analyse the quench-front propagation, the quenching temperature is introduced. In this work, $T_{\min}$ denotes the minimum wall temperature at which film boiling becomes unstable and transitions to liquid contact. When the wall temperature drops to $T_{\min}$, the vapour film collapses, the surface rewets, and the local heat-transfer coefficient increases sharply. This produces a sudden decrease in wall temperature and allows the quench front to advance upward along the bundle.

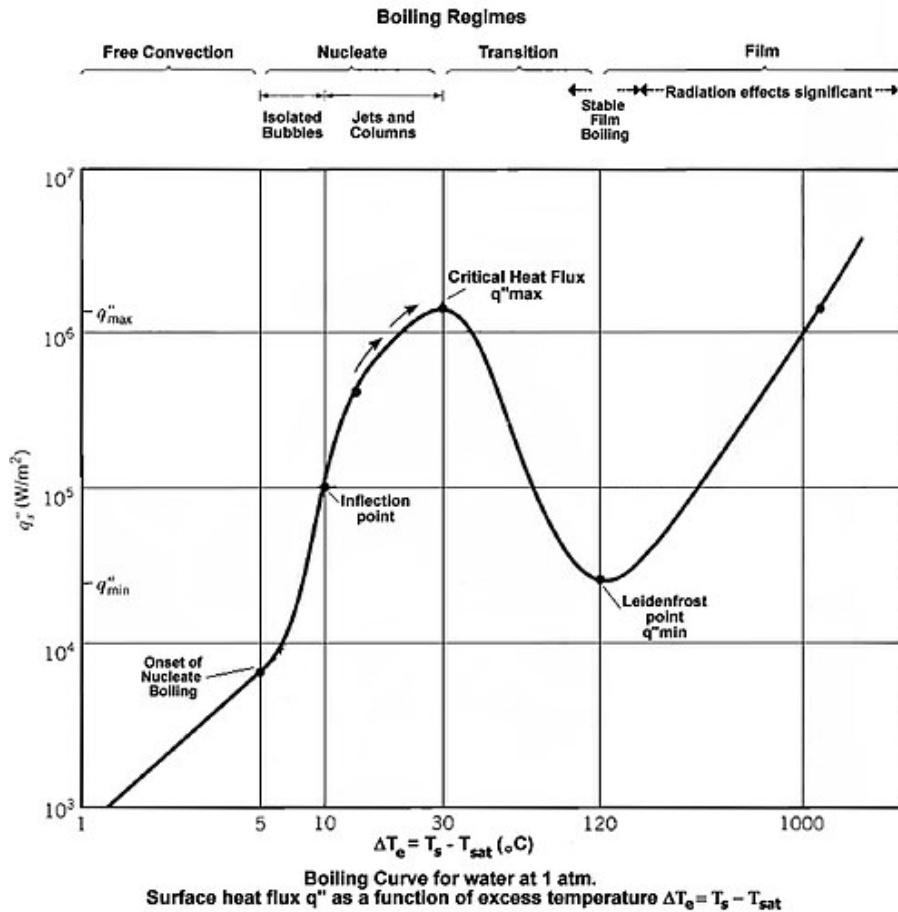


Figure 4: Departure from nucleate boiling curve.

2.2 Existing correlations and TRACE implementation

The current correlations are formulated to be valid within the investigated ranges of pressure, mass flux, wall temperature, and inlet water temperature. However, it was observed that they predict essentially no variation in $T_{\min}$ when these parameters are changed within the intervals

of interest (fig 6, 7, 8, 9). This is contradicted by the experiments that clearly show a noticeable change in quench-front behaviour (fig. 5). Kalinin [1], Lienhard [2], Aursand [3] and Kim [4] correlation were not implemented in TRACE but they were found in the literature. Also this correlation shows a constant value of $T_{\min}$ along the interest intervals of the variables.

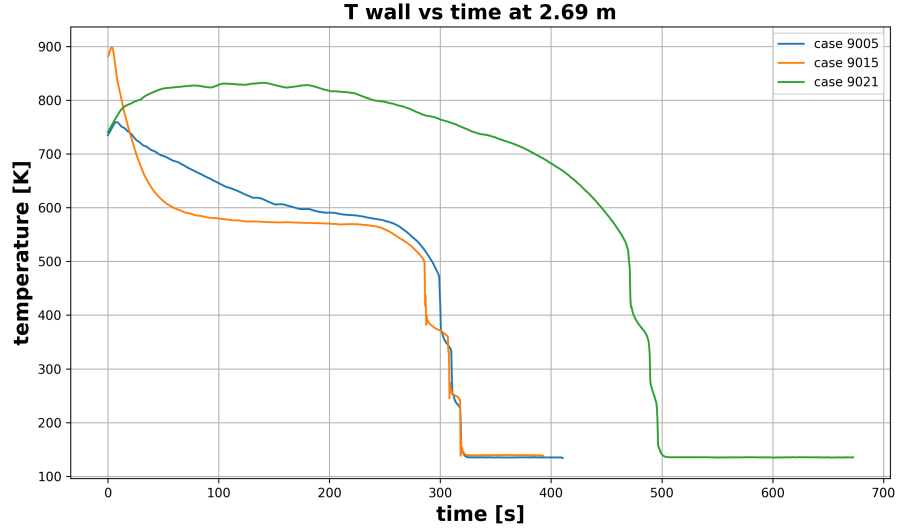


Figure 5: T_{wall} as a function of time for the three cases.

Between case 9005 and 9021 only the mass flow change but it can be see a noteworthy change in behaviour, which can not be justify by the current correlations that show no change in $T_{\min}$ as the mass flow change (fig. 7).

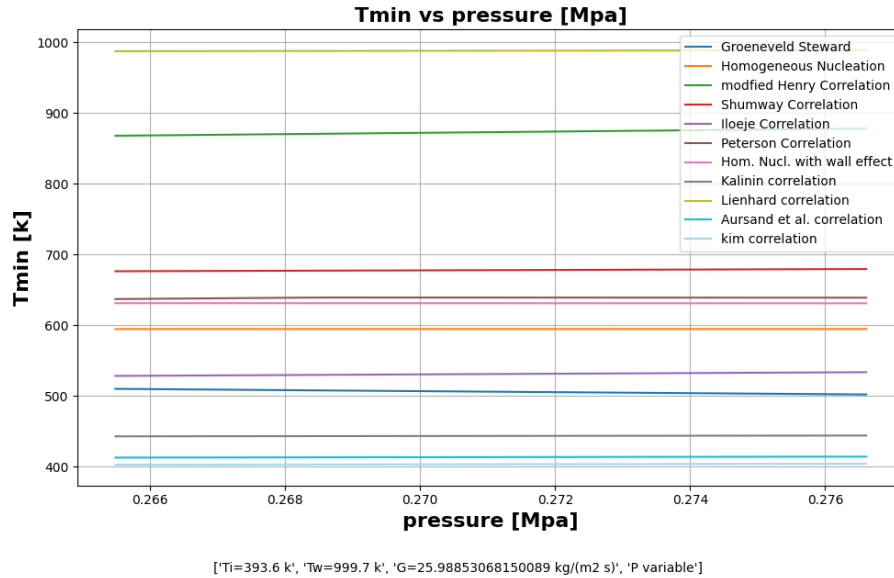


Figure 6: $T_{\min}$ as a function of pressure.

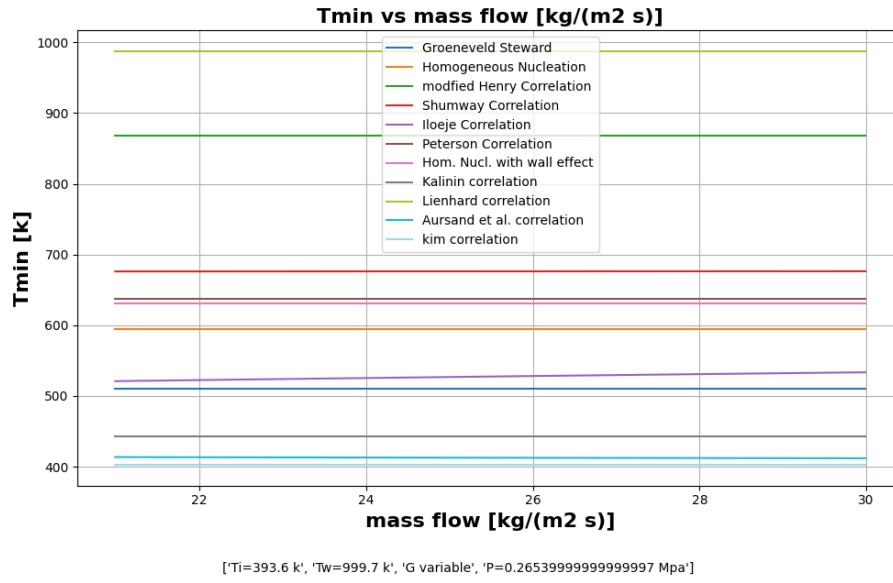


Figure 7: $T_{\min}$ as a function of mass flux.

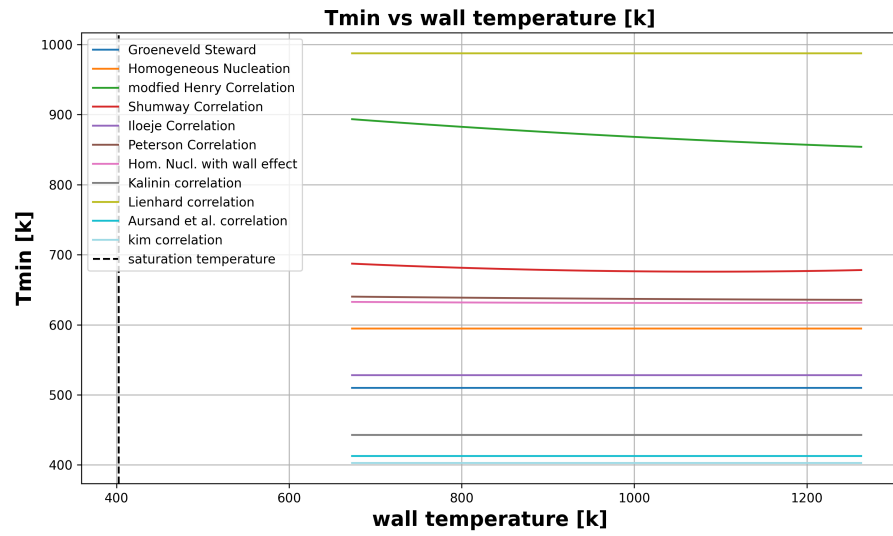


Figure 8: $T_{\min}$ as a function of wall temperature.

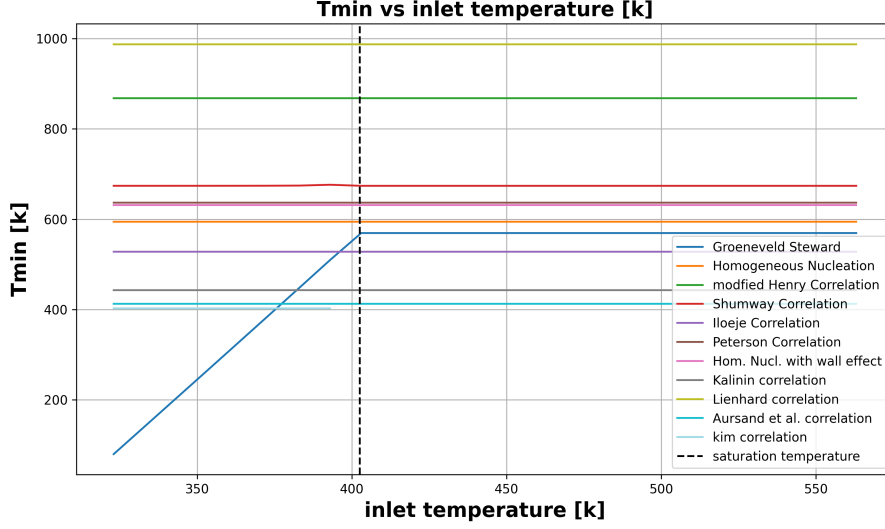


Figure 9: $T_{\min}$ as a function of inlet water temperature.

3 Methodology

3.1 Selection of literature correlations

Additional correlations from the literature were investigated; however, most of them were developed for experimental setups that differ significantly from RBHT reflood conditions, such as horizontal flow configurations, single-surface tests, annular gaps, or narrow rectangular channels. Consequently, their direct applicability to the present rod-bundle reflood experiments is limited. Furthermore as it was shown in the figures 6,7,8,9 the correlations currently tested here exhibit only a mild variation of $T_{\min}$ when the input parameters are changed. This weak sensitivity does not justify the markedly different quench-front behaviour observed among the RBHT cases.

3.2 Development of new correlations

To find a new correlations it was necessary to have a data set of $T_{\min}$ as function of pressure which is the most important parameter.

From TRACE the pressure behavior at different height was extracted. From these data the average pressure after the transient was calculated: to compute a representative mean pressure, the pressure time series $p(t)$ was split into a transient part and a post-transient (quasi-steady) part using a threshold defined from the signal range.

Denoting by $p_{\min}$ and $p_{\max}$ the minimum and maximum values of the measured pressure, the threshold was arbitrarily set as

$$p_{\text{thr}} = p_{\min} + \frac{p_{\max} - p_{\min}}{5}. \quad (1)$$

The last value of time satisfying $p(t) > p_{\text{thr}}$ set the beginning of the post-transient region. The corresponding mean pressures were then computed. Figure 10 shows an example of the pressure evolution at $z_0 = 2.06$ m, highlighting the detected end of transient and the resulting mean static pressure.

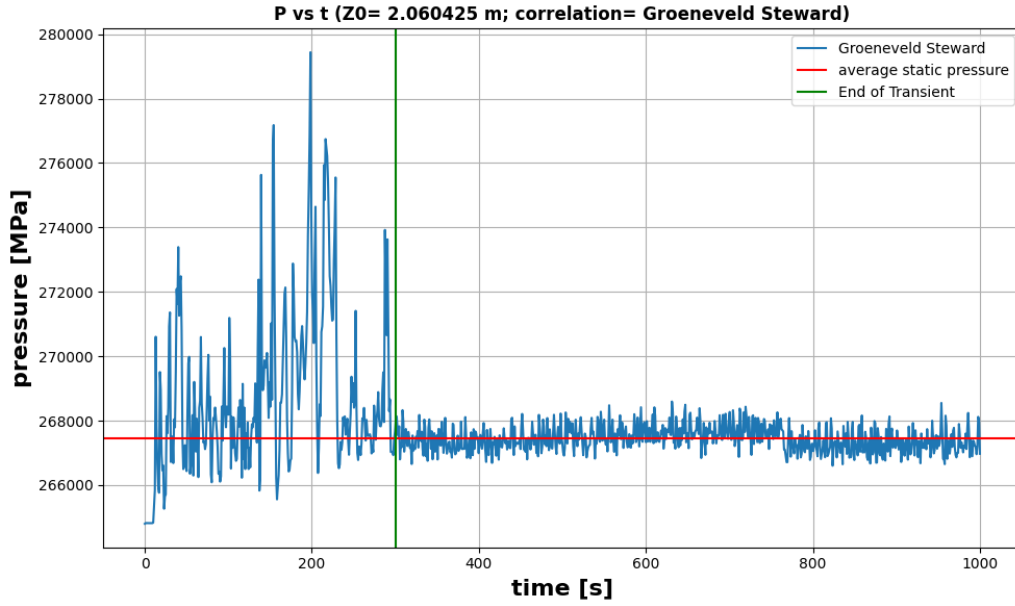


Figure 10: example of pressure as a function of time at $z_0 = 2.06$ m (Groeneveld–Stewart correlation, case 9005). The green vertical line marks the end of the transient t_{end} as defined by the threshold criterion, while the red line indicates the average static pressure $\bar{P}$ computed over the post-transient portion.

Collecting the pressure at different heights for all the correlations it was possible to notice that the differences between the models are negligible: the correlations do not affect the post transient region.

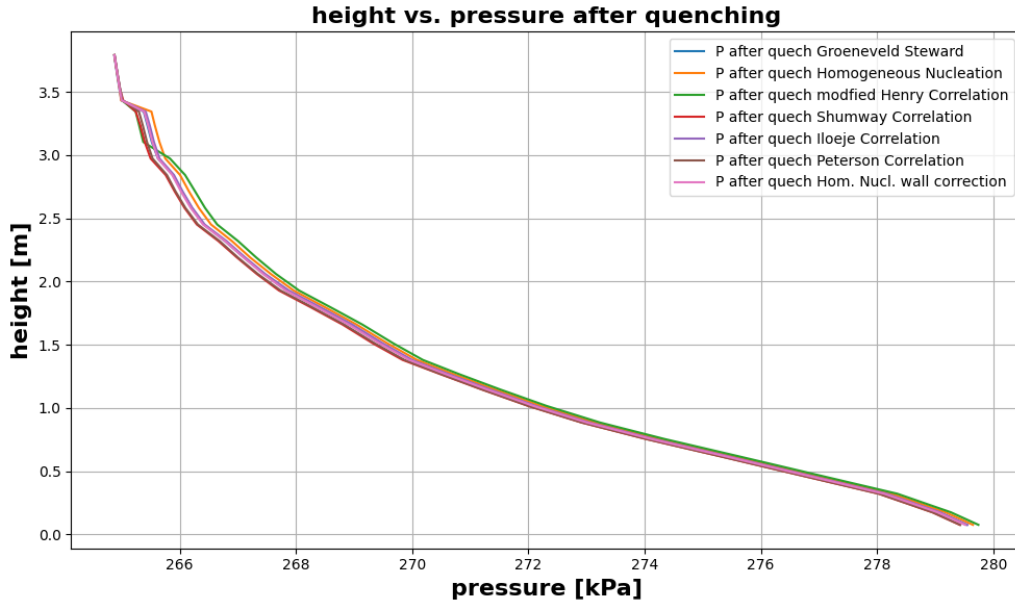


Figure 11: Pressures as a function of height (case 9005).

Afterwards from the documents [5], [6], [7]. the quenching temperature as function of height was extracted for the three cases.

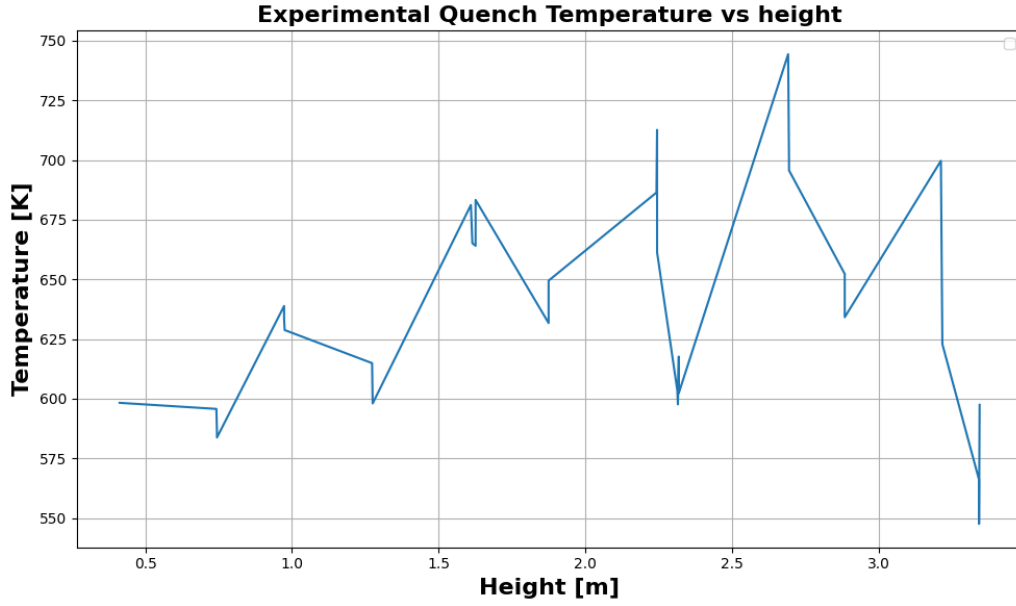


Figure 12: Experimental temperature as a function of height (case 9005).

Knowing the correlation between height and pressure and the correlation between temperature and height it was possible to link quench temperature and pressure. However the experimental data were disturbed by spatial grids distributed along the bundle length so it was necessary to exclude some data.

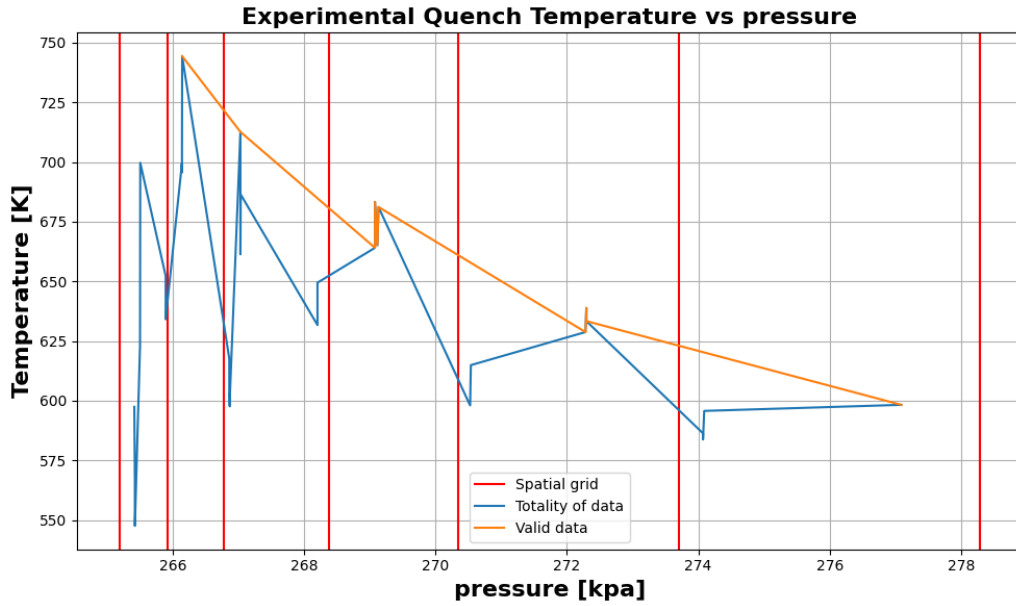


Figure 13: Experimental temperature as a function of pressure, both totality of data and valid data. The pressures corresponding to the positions of the spatial grids are highlighted. (case 9005).

With these data a second order polynomial fit was performed with as parameter pressure and flow velocity. The resulting correlation was:

$$T_{min} = AP^2 + BP + CV^2 + DV + E$$

$$P \text{ [MPa]}, \quad V \text{ [m/s]}$$

$$A = 258983.5784037211$$

$$B = -157495.25394723588$$

$$C = 20467.492873397852$$

$$D = -2279.8625100561435$$

$$E = 24375.421706133824$$

The resulting fitting is shown in the following image:

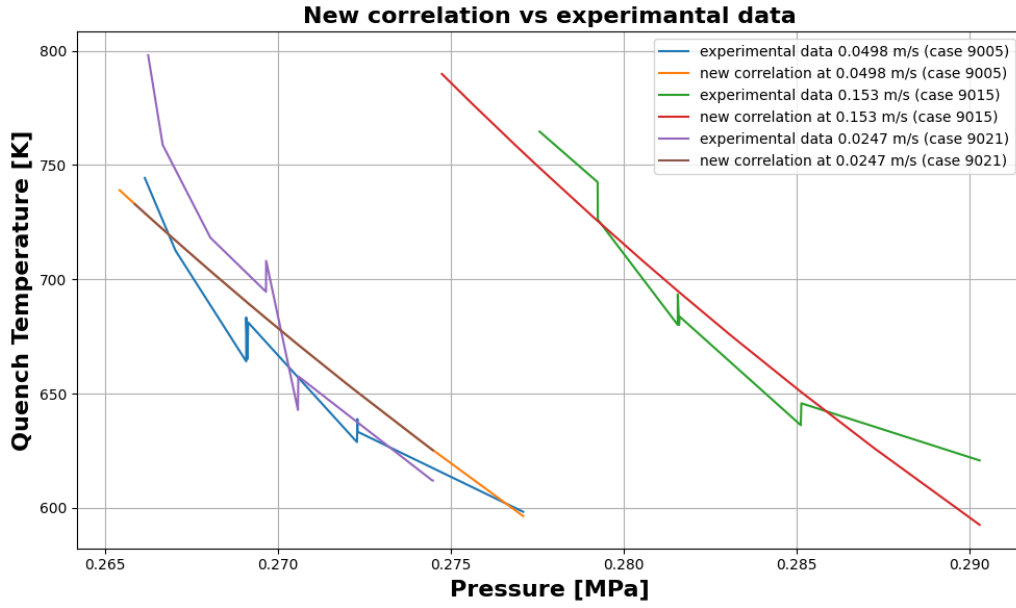


Figure 14: Comparison between predicted values from the new correlation and experimental data for the three cases.

3.3 Validation approach

With the new correlation implemented in TRACE it was demonstrated that the predicted values follow better the different cases. The key point is robustness: individual correlations may excel in one scenario and degrade in another, whereas the new correlation shows consistently good agreement across all cases.

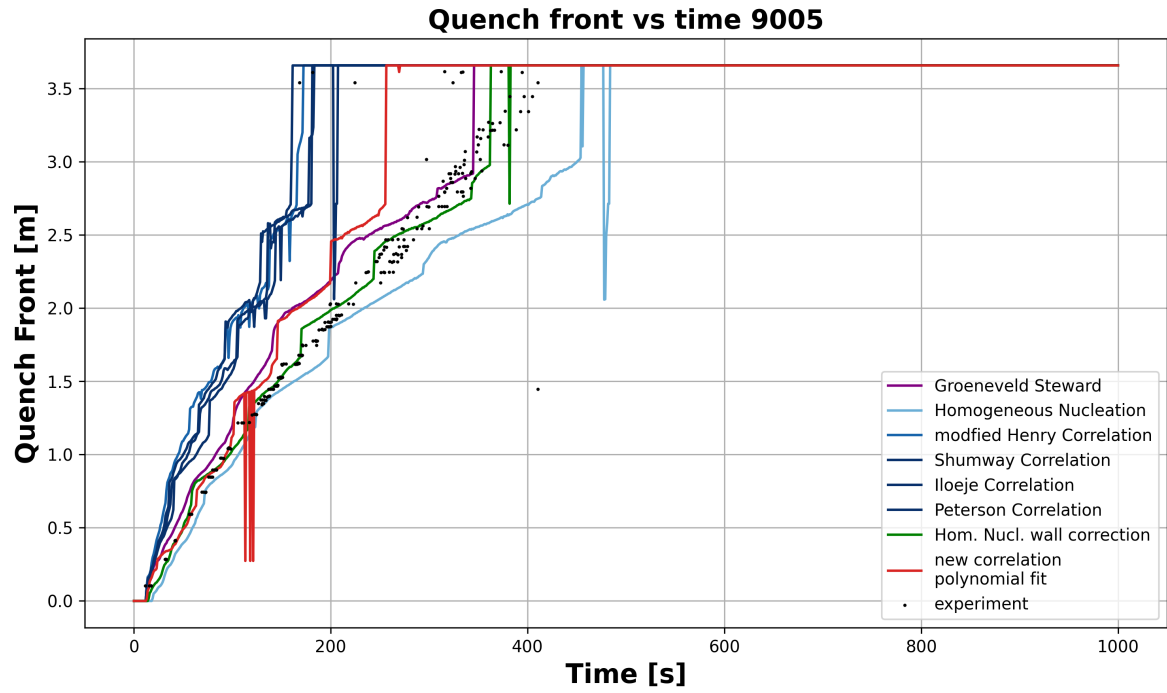


Figure 15: Comparison between experimental data and value calculated with TRACE using different correlations for case **9005**.

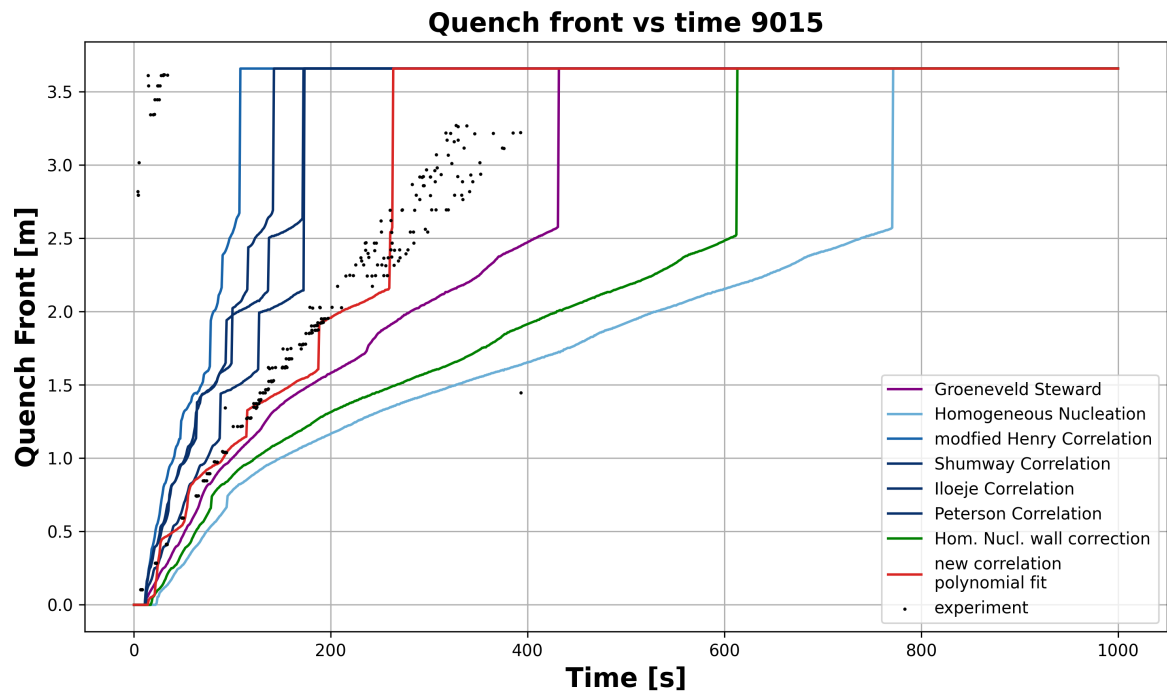


Figure 16: Comparison between experimental data and value calculated with TRACE using different correlations for case **9015**.

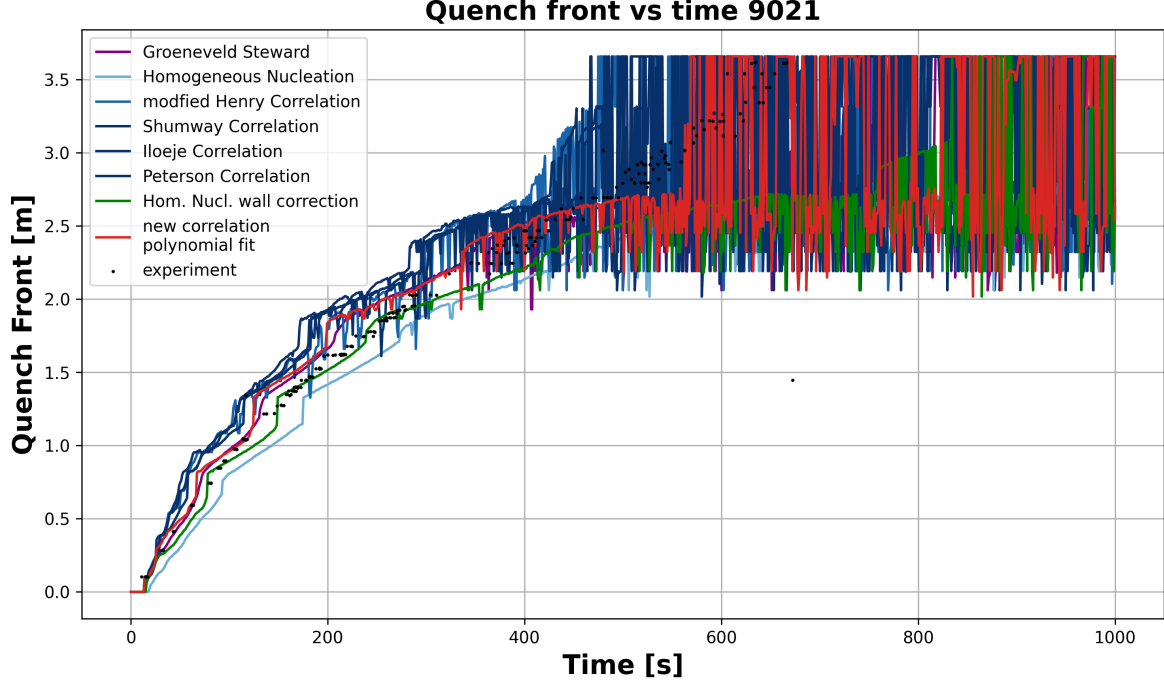


Figure 17: Comparison between experimental data and value calculated with TRACE using different correlations for case **9021**.

4 Conclusion and Future Work

4.1 Conclusions

This project investigated the prediction of the quenching temperature $T_{\min}$ and its impact on quench-front propagation during RBHT reflood experiments simulated with TRACE. The initial comparison highlighted a key limitation: the $T_{\min}$ correlations currently available in TRACE were developed and validated under conditions that differ substantially from the present RBHT tests (e.g. different geometries). As a consequence, their use in RBHT reflood simulations led to inconsistent quench-front predictions, especially when operating conditions were varied.

A second important outcome is that most reference correlations place the dominant emphasis on pressure (and, depending on the model, subcooling/quality), whereas the RBHT cases clearly indicate that the flow rate/velocity (equivalently mass flux) plays a crucial role in determining the quench-front behaviour. In particular, case 9015 (higher mass flux) demonstrates that pressure-driven correlations alone are not sufficient to reproduce the experimental quench-front evolution. This observation motivated the development of a new correlation explicitly including both pressure P and velocity V .

Using RBHT data, $T_{\min}$ values were extracted from experimental measurements and linked to post-transient mean pressures obtained from TRACE pressure fields. After excluding measurements affected by spacer-grid disturbances, a second-order polynomial fit in P and V was derived and implemented in TRACE. Validation against the three dedicated experiments (cases 9005, 9015, and 9021) showed improved agreement with the measured quench-front progression. The key result is robustness: while individual existing correlations may match one case reasonably well but perform poorly in another, the newly derived model provided consistently good predictions across all investigated cases.

An additional insight concerns sensitivity and applicability ranges. In the literature, widely used $T_{\min}$ correlations are often reported as applicable over very broad pressure intervals (e.g. Groeneveld–Stewart is commonly quoted up to ~ 9 MPa, with lower bounds reported around

~ 0.1 MPa). In contrast, the RBHT experiments considered here exhibit noticeably different quench-front behaviours for pressure variations on the order of only a few 10^2 kPa. This highlights that, even within a formally “valid” range, a correlation may still be insufficiently sensitive (or not representative) for a specific facility and reflood configuration. Therefore, facility- and condition-specific tuning of $T_{\min}$ can be necessary for reliable reflood predictions.

4.2 Future work

- Extend the database with additional RBHT experiments and wider parameter ranges to find a more reliable correlation.
- Perform uncertainty quantification and sensitivity studies on the fitted parameters.
- Test portability of the correlation on other facilities of the same type (bundle geometry).
- Finally, the proposed correlation remains empirical and is currently validated only within the RBHT parameter space covered by the three experiments. In a longer project, it would be particularly valuable to expand the model by investigating additional dependencies that are physically relevant for reflood, notably: (i) inlet water temperature / subcooling effects, and (ii) wall temperature (or surface condition) effects, to determine whether they can further improve prediction accuracy and generality without compromising simplicity.

Overall, this study demonstrates that RBHT reflood conditions require a $T_{\min}$ modelling approach that differs from standard TRACE implementations. Furthermore including velocity/mass-flux effects is essential to capture the observed quench-front behaviour which suggest that it could be interesting to implement other parameter as well, such as inlet water temperature and wall temperature.

References

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